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Protocol to accompany paper entitled "Interactions among genes involved in reverse cholesterol transport and in the response to environmental factors in dyslipidemia in subjects from the Xinjiang rural area"

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Abstract

This is a SNP Genotyping protocol

Attachments



[SNP Genotyping proto...](#)

16KB

- 1 **DNA Extraction**Fasting venous blood (200 μ L) was taken from each study subject and a non-centrifugal columnar blood genomic DNA isolation kit (Tiangen, Beijing, China) was used to extract the whole blood genomic DNA. Extracted DNA was verified by gel electrophoresis (0.7% agarose). A NanoDrop spectrophotometer (NanoDrop technologies, Inc.: Wilmington, DE, USA) was used for quantitative determination of DNA concentration and purity: concentration $\geq$ 30 ng/ μ L and purity levels (OD260/OD280) of 1.7–2.0 were considered acceptable. Samples that met these criteria were diluted to 10–30 ng/ μ L using double-distilled water and stored at -80°C until use.
- 2 **PCR Amplification**Primers were designed using the Mysequenom tool (www.mysequenom.com/Home) and Assay Designer 3.0 software (SEQUENOM, Inc.: San Diego, CA, USA). Final PCR reaction volumes were 15 μ L, which included 1 μ L DNA samples, 0.3 μ L dNTPs, 7.4 μ L water, 1.5 μ L 10 $\times$ PCR buffer, 1.5 μ L MgCl_2 , 0.3 μ L Taq enzymes, and 3 μ L PCR amplification primer mixture. Cycling conditions were as follows: predegeneration at 94°C for 4min; followed by 35 cycles of denaturation at 94°C for 20s, annealing at 56°C for 30s, and extension at 72°C for 1min. A final extension step was carried out at 72°C for 3min, after which samples were maintained at 4°C . Reactions were set up in an ice bath and each PCR experiment included a negative control reaction.
- 3 **PCR Products Purification**Shrimp alkaline phosphatase (SAP) was used to remove excess dNTPs from samples after PCR. This step served to ensure the accuracy of single-base extension. The final SAP reaction volumes were 5.0 mL, which included 0.5 μ L 10 $\times$ SAP buffer, 2 μ L PCR product, 2 μ L double-distilled water, and 0.5 μ L SAP enzyme. Reactions were carried out by incubation at 37°C for 40 min, followed by incubation at 85°C for 5 min. The reaction products were stored at 4°C .
- 4 **Single-Base Extension**For single-base extension reactions, final reaction volumes were 6.0 μ L, which included 0.5 μ L Snapshot reagent, 2.5 μ L water, 1 μ L primer mix, 2 μ L purified PCR products. Reaction conditions were as follows: denaturation at 94°C for 30s; followed by 40 cycles of 94°C for 5 s, 52°C for 5s, and finally 52°C for 5s. Reaction products were stored at 4°C .
- 5 **Genotyping Analysis**Take 1 μ L reaction product plus 9 μ L HIDi, 95°C denaturation 3min, immediately ice-water bath, all representative SNP genotyping experiments were done using TaqMan technology on an ABI3730XL system (Applied Biosystems: Carlsbad, CA, USA). T gene-mapper was used to complete the classification and output the results.